

SDM COLLEGE OF ENGINEERING AND TECHNOLOGY, Dharwad-580002

**(An autonomous Institution affiliated to
Visvesvaraya Technological University, Belgaum - 590018)**



Department of Electronics and Communication Engineering

Major Project Phase 1 [18UECL703] Report entitled

“A supply chain for safe distribution of medicines ”

Submitted by

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Students of 7th Semester

Under the guidance of

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Department of Electronics and Communication Engineering

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S.D.M. College of Engineering & Technology
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CERTIFICATE

This is to certify that,

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Students of 7th Semester have satisfactorily completed the Major Project Phase-1 [18UECL703] entitled “A supply chain for safe distribution of medicines” submitted to the Department of Electronics and Communication Engineering, SDM College of Engineering and Technology, Dharwad – 580 002, for the partial fulfillment of the Degree.

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ABSTRACT

This project presents the design and development of an advanced navigation aid system tailored to enhance the mobility and independence of visually impaired individuals. The proposed system integrates cutting-edge technologies such as computer vision, machine learning, and sensor fusion to create a comprehensive solution for real-time navigation and obstacle detection.

The system employs natural language processing and speech synthesis to provide audible feedback to the user. This includes turn-by-turn navigation instructions, information about the surrounding environment, and warnings about obstacles in the path. Additionally, haptic feedback mechanisms are incorporated into the wearable device to further enhance the user's spatial awareness. Vibrations and tactile cues guide the user in the right direction and alert them to the presence of obstacles.

To ensure adaptability and usability, the system is designed to be user-friendly, with customizable settings to accommodate individual preferences and needs. The device can be easily paired with a smartphone application, allowing users to plan routes, set preferences, and access additional features such as GPS integration.

Chapter 1

1.1 Introduction

People with visual impairments face numerous challenges everyday. There are many successful technologies and mobile applications out there today for the visually impaired people to help them in their daily tasks like mobile screen readers, computer screen readers, voice enabled calculators etc., but one of the biggest problems they still face, is navigation and awareness of their immediate environment. India itself is home to 20% of the world's global blind population. According to a survey conducted in the year 2000 the number of visually impaired people in India were 18.7 million. The leading cause of visual impairment was stated as cataract, which alone accounted for 9.5 million visually impaired people. We know for a fact that cataract is curable with surgery, and hence is not the leading cause of visual impairment in the US.

The large number of cataract cases in India can be accounted to the fact that 80% of the visually impaired population, reside in villages and probably cannot afford cataract surgeries. Currently, India has 232 million internet users and 65% of the traffic is from mobile phones. With lowering costs of smart-phones and also data plans, and with local initiatives like Digital India and global initiatives like Facebook's Internet.org, the number of smart-phone internet users in rural sector is increasing at a rapid rate.

A navigation assistive technology which takes advantage of these facts is what we have implemented and proposed in this paper. Visually impaired people use a cane to help them detect the immediate environment ahead like a speed bump, or a puddle etc. They can and do move from place to place. They go to work, walk to the shop etc. Over time, they learn routes to their workplace and other landmarks and can travel alone. The tasks where the difficulty arises are tasks like crossing the street. They also face many dangers of fast approaching vehicles while walking along the road. Many of the visually impaired people use a guide dog. This helps them navigate but they are still largely unaware of the environment and obstacles around them. Many systems for obstacle detection have been proposed like proximity and ultrasonic sensors in the cane, head mounted devices, using a depth sensor like Microsoft Kinect and sonification etc. All of these methods require extra expense or heavy hardware. Even if the hardware is not expensive, the availability of the hardware required is another issue. Apart from this, social stigma such as not wanting to wear a head mount all the time or not wanting to carry an external hardware would result in unpopularity of these devices. These hardware devices usually have a separate external battery which might be inconvenient to carry

and would also require regular replacement. The system we have proposed, requires no extra hardware or cost. It runs on the users smart phone and only requires the camera stream and an internet connection to stream the frames to an offline server for computations. Using socket communication, the results can be sent back in real time. Once the phone is placed in a convenient position, such as in the users shirt pocket such that the camera is looking ahead, or is worn around the neck using a band, the user instantly starts receiving information about the obstacle approaching, what it is, its direction of approach, and also approaching friends/family members. With this information, the user is now more aware of his surroundings and can take appropriate measures.

The user receives only important notifications such as that of approaching or looming obstacles and not that of receding or obstacles far away from him. The user also will not get notifications if an obstacle is moving sideways in which case it is no harm to the user. The system does not give false notifications when the user turns and now faces a new scene due to gating and recalibration after turning. Obstacles are segmented out and are classified into Car, Human, Dog, Chair or Door with the direction of approach.

The system can also recognize faces using a standard face recognition algorithm and this would notify the user when his friends or family are approaching. A majority of the visually impaired people have low vision and are not completely blind. Low vision could result in loss of central or peripheral vision or overall blurriness. A system like this would help them identify faces, obstacles, help them stay out of danger, and help them in navigation in a convenient zero cost manner.

1.2 Motivation

The main motive of this project is to provide an assistive system to blind people which will make them comfortable to walk in any unfamiliar environments. We are going to develop a system that works efficiently well in outdoor. Current navigation device for the visually impaired focus on travelling from one location to another. This focuses on designing a device for visually impaired people that help them to travelling independently also it must be easy to use. The proposed device is used for guiding people who are blind or partially sighted. The device is used to help visually impair people to move with the same ease and confidence as a sighted people. The device is linked with a GPS to identify the location of the blind person.

1.3 Project Scope

This project will explore current technology, specifically the Life pilot GPS devices, as a promising aid for both support and encouragement to the blind and partially sighted as they struggle for an independent life. It will focus on identifying the features that should be included in such a device, as well the feasibility of having that technology succeed in the market. GPS/GSM tracking system are used to track person or vehicle in many systems. It has been found that all these tracking systems are costly than the proposed system. So the proposed system will overcome all the problems, and people can use this system in their home or companies to track their location on Google maps as specific software as the cost of the system is too low.

1.4 Applications

1. Navigation and Route Planning: Assist in navigating unfamiliar places by providing step-by-step directions, helping users reach their destination efficiently.
2. Obstacle Detection: Warn users about obstacles or hazards in their path, using GPS and other sensors to identify and communicate potential obstacles.
3. Location Identification: Help users identify and explore their current location, providing information about nearby points of interest, businesses, or public facilities.
4. Public Transportation Assistance: Guide users through public transportation systems, including bus stops, train stations, and subway routes.
5. Indoor Navigation: Extend navigation assistance to indoor spaces like malls, airports, and public buildings, improving accessibility in complex environments.
6. Crossing Assistance: Provide information about crosswalks, traffic signals, and pedestrian crossings, ensuring safe navigation through intersections.
7. Geotagged Reminders: Set location-based reminders, enabling users to receive notifications when approaching specific places, such as a grocery store or a friend's house.
8. Emergency Assistance: In case of emergencies, the system can guide users to the nearest exit, emergency services, or safe locations.

Chapter 2

Literature Survey

Amutha B and Ponnavaikko M (2009), “Location Update Accuracy in Human Tracking System Using Zigbee Modules”, International Journal of Computer Science and Information Security (IJCSIS), Vol. 6, No. 2.

A location and tracking system becomes very important to our future world of pervasive computing. An algorithm for accurate location information is being incorporated in the human walking model and in the blind human walking model. We want to implement an accurate location tracking mechanism using Zigbee along with GPS, we have incorporated Markov chain algorithm for establishing accuracy.

GPS Technology to Aid the Blind and Partially Sighted in Copenhagen, May 7, 2007, Denmark.

In 2007, Copenhagen pioneered the integration of GPS technology to empower individuals with visual impairments, fostering a more inclusive urban environment. This initiative aimed to enhance independent navigation for the blind and partially sighted population. The implemented system utilized specialized devices equipped with GPS receivers to provide real-time location information. Users received auditory cues and haptic feedback, enabling them to navigate through the city along predefined routes. This marked a significant step towards breaking down barriers and promoting accessibility.

Jafnie Evangeline J (2014), “Guide Systems for the Blind Pedestrian Positioning and Artificial Vision”, Vol. 1, No. 3.

The paper proposes a novel approach to guide system for the blind pedestrian positioning and artificial vision. The proposed system fuses GPS and vision-based positioning to provide accurate and reliable location information to blind pedestrians. The system also incorporates a multi-sensor data fusion algorithm to provide real-time obstacle detection and avoidance.

"A Survey on Talking Assistance about Location Finding both Indoor and Outdoor for Blind People" by N. Nandhini, Vinoth Chakkaravarthy, G. Deepa Priya, and K. Santha Sheela.

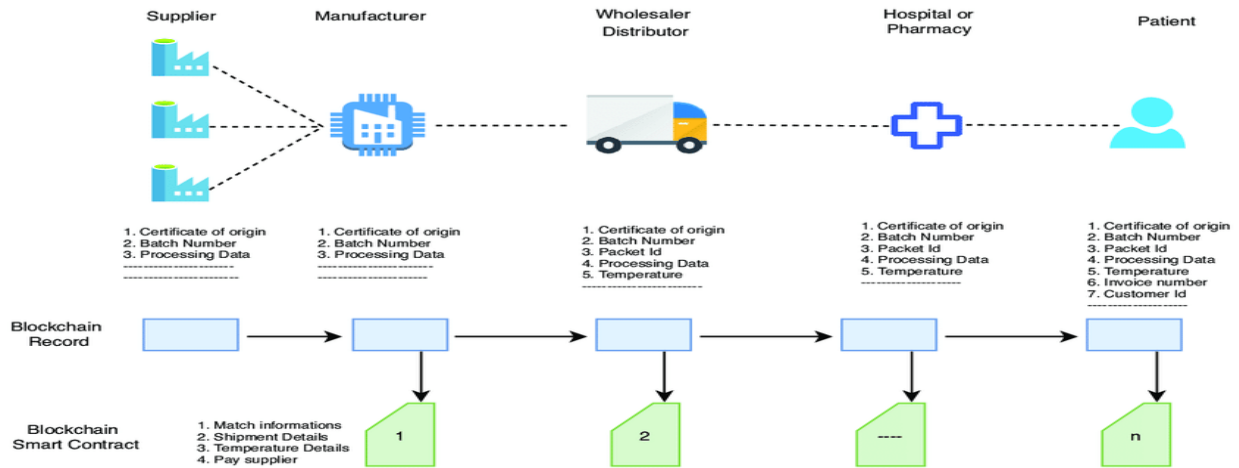
The paper provides a comprehensive overview of talking assistance systems for location finding for blind people. The authors discuss the various types of talking assistance systems, including indoor navigation systems, outdoor navigation systems, and wearable systems. They also discuss the challenges of developing effective talking assistance systems, such as the need for accurate location information and the need for a natural and intuitive user interface.

"Assistive Technologies for Visually Impaired People" by Sándor Tihamér Brassai, László Bako, and Lajos Losonczi.

The paper provides an overview of assistive technologies for visually impaired people. The authors discuss the various types of assistive technologies, including devices for reading and writing, devices for mobility, and devices for communication. They also discuss the challenges of using assistive technologies, such as the need for training and the need for accessible infrastructure.

Chapter 3

Methodology



1. Owner deploys the smart contracts to the Ethereum Blockchain.
2. The Owner authenticates and registers the entities of the chain.
3. Supplier registers a new Raw Material.
4. Raw Material Contract is deployed for the newly created raw material.
5. Corresponding Transaction Contract is also deployed for the newly created raw material.
6. Raw material registered successfully
7. Supplier transfers the raw material to the Transporter.
8. Supplier updates the product status and creates a Transaction in the Transaction Contract.
9. Transporter transfers the raw material to the manufacturer.
10. Manufacturer verifies the source of the raw material.
11. Manufacturer updates the product status and creates a Transaction in the Transaction Contract.
12. Manufacturer registers a new medicine.
13. Medicine Contract is deployed for the newly created medicine.
14. Corresponding Transaction Contract is also deployed for the newly created medicine.
15. Medicine registered successfully.
16. Manufacturer transfers the raw material to the Transporter.
17. Manufacturer updates the product status and creates a Transaction in the Transaction Contract.
18. Transporter transfers the raw material to the Wholesaler.
19. Wholesaler verifies the source of the raw material.
20. Wholesaler updates the product status and creates a Transaction in the Transaction Contract.
21. Wholesaler transfers the raw material to the Transporter.
22. Wholesaler updates the product status and creates a Transaction in the Transaction Contract.

23. Transporter transfers the raw material to the Distributor.
24. Distributor verifies the source of the raw material.
25. Distributor updates the product status and creates a Transaction in the Transaction Contract.
26. Distributor transfers the raw material to the Transporter.
27. Distributor updates the product status and creates a Transaction in the Transaction Contract.
28. Distributor transfers the raw material to the Customer.
29. Customer verifies the source of the medicine through a temporary node.
30. Customer updates the status of the product through a temporary node.

Chapter 4

Components:



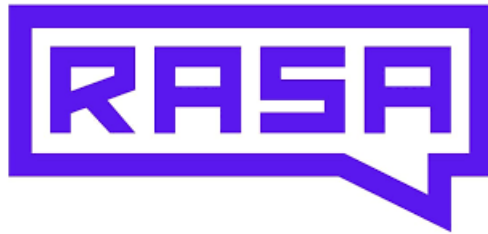
Blockchain is a new technology that is gradually emerging with the increasing popularity of digital currencies such as Bitcoin. It is essentially a distributed ledger database. Blockchain records transactions that have occurred by establishing a database maintained by all network nodes and the entire process is open, transparent, and irreversible. According to the participants, it can be divided into the public blockchain, consortium blockchain, and private blockchain, and the consortium blockchain and the private blockchain are collectively referred to as the permissioned blockchain. The public blockchain is completely decentralized in the real sense. At any time, any node will join or leave the decision to build a new block. In the permissioned blockchain, though, the decision to build a new block is made by certain trusted nodes. It has been applied to copyright management, identity authentication, and data storage.



Ganache

Ganache is used for setting up a personal Ethereum Blockchain for testing your Solidity contracts. It provides more features when compared to Remix. You will learn about the features when you work out with Ganache. Before you begin using Ganache, you must first download and install the Blockchain on your local machine.

RASA CHATBOX



Rasa is an open-source machine learning framework for automated text and voice-based conversations. Rasa can be used to understand messages, hold conversations, and connect to messaging channels and APIs. Following are the steps of how an assistant built with Rasa responds to a message:

1. The received message is routed into the interpreter, which translates it into a dictionary of the original text, intent, and entities discovered. This is handled by NLU.
2. The tracker object records the current state of the conversation. It receives information that a new message has come in.
3. The policy receives the tracker's current state.
4. The policy determines what action to do next.

Chapter 5

Future Work

Future work will focus on integrating haptic feedback and sensory enhancement, enabling users to perceive their surroundings through multiple sensory modalities. Adaptive learning and collaborative navigation features will further enhance the user experience, providing personalized guidance and fostering social connections. Open-source development and universal design principles will promote accessibility and innovation, ensuring that navigation aids reach a wider audience and address the diverse needs of visually impaired individuals. As these advancements unfold, the future of navigation aids looks bright, promising a world where everyone can navigate their surroundings with ease and confidence.

Conclusion

Navigation aids for visually impaired individuals have undergone significant advancements, and the future holds even more exciting possibilities. By harnessing the power of artificial intelligence, real-time mapping, and personalized feedback, navigation aids can empower .Enhanced object detection and identification will enable navigation aids to provide more detailed information about the environment, while smart indoor navigation will facilitate seamless movement within buildings. Personalized user interfaces and integration with wearable technology will cater to individual needs and preferences, ensuring an optimal user experience.

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4. "A Survey on Talking Assistance about Location Finding both Indoor and Outdoor for Blind People" by N. Nandhini, Vinoth Chakkaravarthy, G. Deepa Priya, and K. Santha Sheela.
5. "Assistive Technologies for Visually Impaired People" by Sándor Tihamér Brassai, László Bako, and Lajos Los

